

VAISHNAVI

(+91-8826337998) | gvaishnavi00@gmail.com | [linkedin.com/in/gvaishnavi21](https://www.linkedin.com/in/gvaishnavi21) | **Skype:** live:.cid.895201219c1975f6

EDUCATION

Indian Institute of Technology Bombay | *CPI-9.49/10.00* (2022-2024)
M.Sc. Environmental Science & Engineering (Department Rank: 1) Mumbai, India
• **Key Courses:** Envi. Management, Circular Economy, Env. Law & Policy, Solid Waste Management, Energy Conversion, Membrane Processes, Municipal Water & Wastewater, Biological & Physicochemical Treatment

Banaras Hindu University | *CPI-9.47/10.00* (2019-2022)
B.Sc. (Hons.) Botany (Double Gold Medalist) (Rank: 1/110 Students) Varanasi, India

PROFESSIONAL EXPERIENCE

IITB Green Energy & Sustainability Hub | *Senior Associate (Training & Edu.) (Jan'25-Present)*
Skills: Project Management, Stakeholder Engagement, Sustainability Communication and Mentorship
• **Mentored 25 Fellows**, managed **15 projects**, developed **10+ reports & proposals**, conducted **25+ masterclasses & immersion activities** and led two Cohorts of Living Lab Fellowship Program for Campus Sustainability
• **Facilitated 200+ hours** of training program for working professionals & summer school for **40+ students**
• Led engagement activities and project presentations at CSR networking events and developed CSR Proposals

Global Energy & Environmental Engr. Services (GE3S) | *ESG Assistant (May'24-Aug'24)*
Skills: Carbon Accounting and Mitigation, Business Development, Decarbonization
• Developed the decarbonization plan for clients to achieve **6 % reduction** in **GHG emissions** by 2030
• Designed the sector-wise **GHG calculation tool** by benchmarking different KPIs for **Scope 1,2,3 emissions**
• Created the **business plan** for investment in greenhouse technology by analysing market trends & **ROI**

Life Link Eco Technology Pvt. Ltd. | *Research & Development Trainee (Mar'24-Jun'24)*
Skills: Consultancy, Research & Data Analysis
• **Documented 50+ plants** morphology/adaptability & studied its role in bioremediation of **pollutants** in stormwater.
• **Researched & Calculated** the GHG emission from wastewater during transportation & onsite treatments

Water Resource Department| Government of Maharastra | *Project Intern (May'23- May'23)*
Awarded **Letter of Recommendation** for a commendable contribution towards the pilot study
• Reported & studied status of water bodies through **10+ field visits**, **50+ stakeholder interactions** & surveys
• **Analysed the 5 physicochemical parameters** of river for **water quality monitoring** & **executed compliance**

RESEARCH EXPERIENCE

Extraction & Characterization of Plant-Based Biosurfactants | *MS Thesis (Jan'23-Apr'24)*
Guide: Prof. Suparna Mukherji, ESED, IIT Bombay | **Awarded AA (10 CGPA) in master's thesis**
• Researched **sustainable** alternatives for synthetic surfactants, extracted surfactants from aloe vera waste & reetha
• Experimented with plants to extract and characterise surfactants and explore their application in **solubilisation** of **petroleum hydrocarbons** such as PAHs for **surfactant-enhanced bioremediation** of wastewater

Metaproteomics Analysis of COVID-19 Patients | *Research Intern (Jul'21- Feb'22)*
Guide: Prof. Sanjeeva Srivastava, BSBE, IIT-Bombay | **Awarded letter of Recommendation**
• **Visualized mass spectrometry (MS)** data & performed **spectral analysis** for **5000+** peptide sequences & carried out **sequence alignment** to identify co-infecting microbial peptides in nasopharyngeal swab samples
• **Documented** data from **70+** journals on secondary infection in Covid patients & **examined** their global trends

KEY PROJECTS

GIS-Based Spatial & Temporal Groundwater Quality Assessment | (Jan'23-Apr'24)
Guide: Prof Anil K Dikshit & Prof. Srinidhi Balasubhramaniam **Tools used:** ArcGIS, ArcMap, MS Excel
• **Calculated the groundwater quality index** of Delhi to understand the water quality for site-specific treatment
• **Analyzed the spatial-temporal variation** of Delhi's groundwater quality for **three years** using **ArcGIS**

Lifecycle Analysis of Green and Sustainable Foods by Haldiram | (Aug'23-Nov'23)

Guide: Prof. Srinidhi Balasubramaniam, ESED, IIT Bombay **Tools used:** OpenLCA & MS Excel

- **Measured** environmental impacts of 'wheat flour halwa' to **analyse** its potential as green & sustainable food
- **Suggested** methods to reduce the impact of the product by **20%** based on the interpretation of the **Open-LCA**

Circular Economy Principles in Plastic Waste Management of Campus | (Jan'23-Apr'23)

Guide: Prof A. Deshpande & Prof P. Kalbar, CUSE, IIT Bombay **Skills:** Problem-solving, Frameworks

- Conducted **market research** to determine the consumption pattern of beverage bottles at canteens in the IIT-B
- **Devised** the **framework** for plastic-waste management through EPR, innovation & circular economy principles

Algal Diversity Identification from Different Sites on the BHU Campus | (Jun'21-Dec'21)

Guide: Prof. Prashant Singh, BHU **Skills:** Microscopy, Microbial Sampling & Reporting

- **Collected** and **purified** algal species from the BHU campus and differentiated them based on their morphology
- Documented **30+** algal species at BHU campus, along with their habitat, ecological importance, & morphology

Biodiversity Study on Angiospermic Plants at BHU Campus | (Jun'21- Dec'21)

Guide: Prof. Abhishek Dwivedy, BHU **Skills:** Herbarium making, Reporting & Research

- **Collected 30+** Angiospermic plant species & **documented** along with morphological features in a herbarium
- Extensively **studied** medicinal properties & economic importance of **50+** plant species & **documented** them.

LEADERSHIP & MANAGEMENT EXPERIENCE

Treasurer | Environmental Science and Engineering Association, ESEA (Jan'23-Jun'24)

- **Spearheaded** Alumni & Industrial Connect series to mentor **300+** **students** for career in **Env. & Sustainability**
- **Managed finances** & **Conducted 20+** workshops, engaged **700+** students & increased membership by **30%**

Core Team Member- Project & Policies | Sustainability Cell (Jan'23-Dec'23)

- **Established first** interhostel sustainability competition, which ranks hostels using **benchmarked KPIs & Index**
- **Devised** and formulated IITB's **First Sustainability Report**, reporting impact across **17 UN SDGs**

CONFERENCES & WORKSHOPS

- Presented paper on "**Extraction & Characterization of Biosurfactant from Sapindus mukorossi by Ultrasound-Assisted Extraction (UAE)**" at **National Environment Conference** held at **IIT Bombay**
- **Proposed** the solution by different stakeholders during the breakout session at the **National Solid Waste Conference 2023**, organized by the National Solid Waste Association of India (NSWAI)

VOLUNTEERING EXPERIENCE

- Core Member of **Bombay Bookies** (India's Largest Book Reading club) founded by **Shantanu Naidu**, volunteered book collection and donation drive and helped in community engagement for weekly reading session
- Associated with **Walk for Plastic** for clean-ups, plastic collection & funding the education of poor children
- Member of **Rotary International** from Sep '14 - Sep '17. Assigned responsibilities like **Vice president** ('15-'16) & **treasurer** ('16-'17). Headed the Adult Literacy Program, Blood Donation, & Tree Plantation Drive

LABORATORY & TEACHING EXPERIENCE

Teaching Assistant | 12 Months (Aug'23-Dec'23, Aug'24-Dec'24)

- **Mentored** a class of **30+** students as a Teaching Assistant (TA) in the **Env. Chemistry Laboratory** (2023) and **Env. Microbio. & Ecology** (2024) course, prepared assignments/quizzes and evaluated exams
- **Experiments:** Water quality monitoring by BOD, COD, TSS, Turbidity, Nitrates, Sulphates and Chlorides test

Laboratory Courses

Water Quality Monitoring	Tests such as Alkalinity, Hardness, BOD, COD, TOC, Ions Test and Titration
Microbiology	Extraction Techniques, Bacterial Culturing & Staining, MPN, Membrane Filtration
MISC	Microscopy, FTIR, UV-Vis Spectroscopy, HPLC, PCR, LC/MS

EXTRACURRICULARS

Languages	English (TOEFL ibt: 98/C1 (R-25/C2, L-29/C2, S-23/C1, W-21/C1), French (Beginner)
Technical	Arc GIS, OpenLCA, Matlab, MS Office, EPANET, AutoCAD, Visual Minteq, Python
Awards	Honourable Mention Certificate for research, diplomacy, & oratory skill in the "E-Youth Parliament", Secured second position in International Essay Writing (Jun 2020)